J.P.Morgan

Global Markets Strategy
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U.S. Bond Futures Rollover Outlook

September 2026 / December 2026

US Rates Strategy
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- Repo rates have been volatile in recent months due to the inflation concerns associated with the Middle East conflict, Fed credibility issues after the last FOMC press conference, and the downside July payrolls surprise shifting Fed expectations. Current repo pricing is slightly more hawkish than our economists' forecast, and repos could thus be biased lower. However, with the inflation print behind us, they are unlikely to materially change until the next payroll print and are thus likely to be stable over the course of the roll
- Aggregate open interest across the Treasury futures complex has continued to rise and set fresh all-time highs. Asset manager accounts are net long in all sectors, with positioning near all-time highs in TY. Asset manager longs have declined in the US and WN sectors over the past few weeks and are not as elevated with respect to history. We expect technical imbalances to exert bearish pressure on the weighted TY calendar spreads, to exert mildly bearish pressure on the WN and UXY calendar spreads, and to not be a driver in other sectors
- All contracts except US and WN have different front and back CTDs, making calendar spreads directional with yields in those sectors; we thus recommend BPV-neutral hedge ratios for the roll. The CTDs appear to be fairly priced relative to one another, so we don't expect CTD relative value to be a driver of calendar spreads in this roll cycle
- Basis convergence is likely to be a factor in the US and WN sectors, where the back contracts appear to underprice the embedded optionality. Basis convergence could thus exert modest widening pressure on the US and WN calendar spreads
- With all of the CTDs having high coupons and volatility being low relative to earlier in the year, we don't expect the wildcard option to be significant in this roll cycle absent a further selloff in yields or steepening of the curve, in which case it could be relevant for US if the CTD switches to a lower-coupon bond
- Overall, we are neutral on the weighted WN calendar spread due to offsetting pressures, bullish on the weighted US calendar spread due to basis convergence, and bearish and mildly bearish on the TY and UXY weighted calendar spreads, respectively, on the back of investor positioning. We are neutral on the weighted TU and FV calendar spreads

	Front Price	Calendar Spread (/32nds)	HR*	CTD Front	CTD Back	View	Main drivers
WN	110-15	0-12.625/32nds	990	4 Nov '52	4 Nov '52	Neutral	Basis convergence, investor positioning
US	109-07	0-15.625/32nds	980	5 May '45	5 May '45	Bullish	Basis convergence
UXY	110-12	0-09.625/32nds	977	4-1/8 Feb '36	4-3/8 May '36	Mildly bearish	Investor positioning
TY	108-20	0-08.375/32nds	960	4-1/4 May '33	WI Aug '33	Bearish	Investor positioning
FV	106-13.5	0-03.6875/32nds	927	3-1/2 Nov '30	3-1/2 Feb '31	Neutral	N/A
TU	103-00.375	0-03.6875/32nds	872	4-1/8 Jun '28	4-5/8 Sep '28	Neutral	N/A

* Hedge ratio: recommended number of back contracts per 1,000 front contracts. Contract prices and calendar spread levels are based on live levels on 8/12/2026 at 11:30am. Calendar spread levels are quoted at theoretical mids, and front contract prices are quoted at the bid side
Source: J.P. Morgan

See page 10 for analyst certification and important disclosures.

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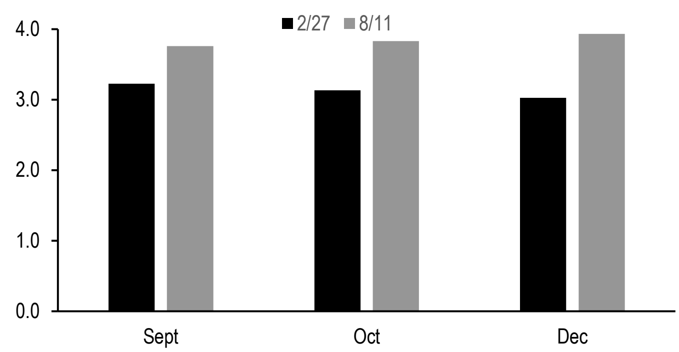
Overview

The September-December Treasury futures roll cycle is upon us, and so much has happened since the last roll cycle. On the geopolitical front, the conflict in the Middle East has continued despite reaching a temporary ceasefire back in June. This backdrop continues to exert pressure on inflation and is one of the primary drivers of the shift in Fed expectations relative to earlier in the year (**Figure 1**). On the monetary policy front, Chair Warsh now has two FOMC meetings and press conferences under his belt, and the FOMC held rates at both of the meetings. However, Warsh's comments in the last press conference on the PCE inflation target and how the balance sheet could be in play to fight inflation steepened the UST curve and were interpreted by markets as a credibility issue. Last Friday, the July payrolls print surprised to the downside, curbing some of the hawkish market pricing. As such, forward financing rates, which can be a key driver of calendar spreads, have been volatile in the past three months, and the rolling 3-week range in repo rates is currently ~16bp (**Figure 2**). With repo rates currently priced more hawkish than our economists' forecast, there is a risk that they could be biased lower if further inflation prints point to softening. However, with the CPI behind us and coming in line with expectations, **we expect financing rates to be mostly stable over the course of the roll and thus not be a driver of calendar spreads.**

Figure 1: Fed expectations have repriced hawkishly since the beginning of the conflict

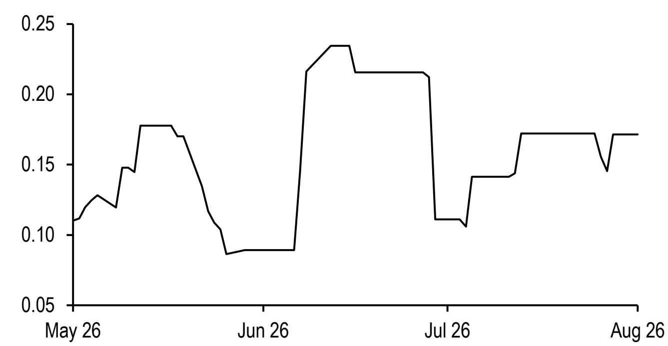
1M forward OIS pricing for Sep, Oct, Dec 2026 FOMC meeting dates as of 2/27 and 8/11; %



Source: J.P. Morgan

Figure 2: Repo rates have been volatile over the past few months, trading in a ~16bp range

Rolling 3-week range for SOFR between 9/30/2026 and 12/30/2026, past 3M; %



Source: J.P. Morgan

One of the consequences of the increase in long-end yields and steepening of the curve was the resurgence of delivery optionality in the longer-dated contracts. We wrote about this in detail previously (*Born(e) in the US(A)*,7/31/2026) and discuss further in CTD switch optionality, basis convergence, and the wildcard option. In summary, **long basis positions in the back contracts of the longer-dated contracts could be attractive for investors who would like to position for a further selloff in yields, which could, all else equal, exert bullish pressure on these calendar spreads.** Additionally, the back basis in these contracts appears to be underpricing the delivery option value; we thus see basis convergence playing a factor in calendar spreads in the US and WN contracts during this roll.

With most of the contracts having high-coupon CTDs, carry is positive across almost all contracts (with the exception of FV, where carry is modestly negative), and they are thus priced to the last delivery day (**Figure 3**). As a result, we expect delivery to be weighted towards later in the delivery month, as has been the case in the past two quarters (**Figure**

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4). While we expect delivery to occur later, we recommend investors who would like to avoid delivery to roll their contracts ahead of the first delivery day. **Another consequence of the CTDs having high coupons and thus high conversion factors, and with volatility being low relative to its highs earlier in the year, is that we don't expect the wildcard option to be a factor in this roll cycle absent a further selloff in yields or steepening of the curve, in which case it could be relevant for US if the CTD switches to a lower-coupon bond.**

Figure 3: Current CTDs have positive carry for all contracts except FV, making the last delivery date optimal for those contracts

Front-contract CTDs, optimal delivery dates, and carry to the optimal delivery date for the respective bonds (ticks); 8/11/2026

Contract	Front Ctr. CTD	Optimal Delivery	Carry
TU	4.125 Jun '28	10/5/2026	1.6
FV	3.5 Nov '30	9/1/2026	-0.3
TY	4.25 May '33	9/30/2026	2.3
UXY	4.125 Feb '36	9/30/2026	2.2
US	5 May '45	9/30/2026	5.5
WN	4 Nov '52	9/30/2026	3.8

Source: J.P. Morgan

Figure 4: In recent roll cycles, delivery has skewed toward the last delivery date in the longer-tenor contracts

Weighted-average number of days from first delivery date for various UST futures contracts for the past 6 roll cycles

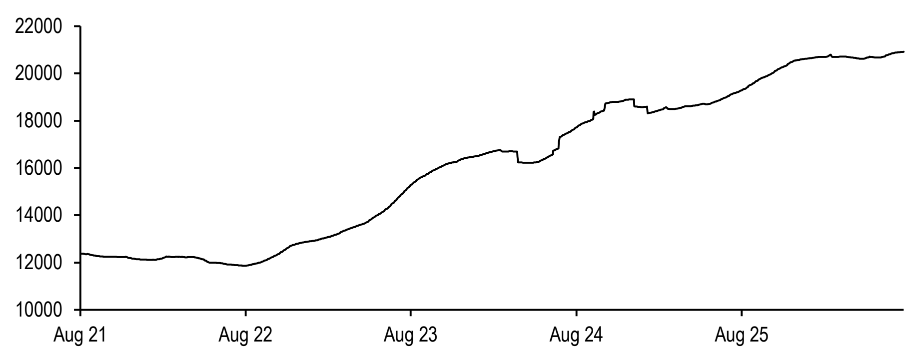
Contract	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26
TU	1	1	19	1	30	13
FV	3	1	1	8	22	11
TY	2	5	3	7	17	29
UXY	0	15	6	15	12	29
US	16	27	28	30	14	29
WN	13	28	23	29	24	27

Source: J.P. Morgan, CME

Over the past few months, aggregate open interest in the Treasury futures complex has continued to increase and is now at new highs (**Figure 5**). **This means that technical imbalances could once again play a factor during this roll cycle**. Our metric to gauge positioning imbalances is net asset manager longs (as they prefer to avoid delivery and roll their positions early). Asset managers are net long across all contracts. Imbalances are very high with respect to recent averages for the **TY contract, where we expect positioning to exert a bearish bias on calendar spreads. In the WN and UXY sectors, we expect positioning to exert a mildly bearish impact on calendar spreads**. **We don't expect positioning to have an impact on the US, FV and TU calendar spreads**.

Figure 5: Open interest across the futures complex has set a new all-time high

Total risk-weighted open interest in Treasury futures contracts*, 000s of UXY equivalents; past 5Y



* Calculated as the weighted sum of the 3M moving average of the open interest across the TU, FV, TY, UXY, US, and WN contracts using weights of 0.5, 0.5, 0.75, 1, 1.5, and 2.5, respectively
Source: J.P. Morgan

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All contracts except for WN and US have different CTDs into the front and back contracts. However, we don't see any relative value opportunities between the CTDs. **With the different CTDs, we recommend BPV-weighting the contracts when rolling contracts forward**. For our recommended hedge ratios, please see Contract tables.

In summary, **we are neutral on the weighted WN calendar spread**. The mildly bearish pressure from the elevated investor positioning is modestly offset by the mildly bullish pressure from the convergence of both the front and back bases. **We are bullish on the weighted US calendar spread owing to basis convergence**. **We are mildly bearish and bearish on the weighted UXY and TY calendar spreads, respectively, due to investor positioning**. **We are neutral on the weighted FV and TU calendar spreads**.

Figure 6: Summary of our views on calendar spreads and their main drivers

Main drivers of calendar spreads and their impacts on the calendar spreads

	WN	US	UXY	TY	FV	TU
Basis Convergence	Mildly bullish	Bullish	-	-	-	-
Investor Positioning	Mildly bearish	-	Mildly Bearish	Bearish	-	-
CTD Relative Value	-	-	-	-	-	-
Fwd. Financing Rates	-	-	-	-	-	-
Conclusion	**Neutral**	**Bullish**	**Mildly Bearish**	**Bearish**	**Neutral**	**Neutral**

Source: J.P. Morgan

Investor positioning

Investor positioning will likely be a factor during this roll cycle because aggregate open interest in the futures complex has continued to rise and has reached a new all-time high (Figure 5). This means that large gross positions must be rolled within a short window of time, which can exacerbate technical imbalances. To gauge such positional imbalances, we look at the net asset manager open interest in each contract. Asset manager accounts in aggregate have a much stronger preference to avoid delivery risk (usually as policy), which forces them to roll their longs early. Therefore, **sectors in which asset manager accounts are significantly net long tend to see narrowing pressure on the calendar spread**.

Asset manager accounts are net long in all sectors (**Figure 7**). Positioning is most stretched in TY, which sits near its all-time high (**Figure 8**). In the WN sector, net longs had been growing steadily over the course of the year but have declined in the past couple of weeks, likely reflecting the concerns around long-end yields. Similarly, net longs have declined in US over the past couple of weeks, likely reflecting concerns with the long end as well as the increased optionality in these long-dated contracts. In contrast, positioning in the TU and FV sectors is currently sitting near the middle of their historical ranges, after declining significantly in the past few months (**Figure 9**). Therefore, **we expect investor positioning to exert bearish pressure on the TY calendar spread, mildly bearish pressure on the WN and UXY calendar spreads, and no pressure on the US, FV and TU calendar spreads.**

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Figure 7: Asset manager accounts are net long in all sectors, with positioning particularly elevated in the WN and TY sectors

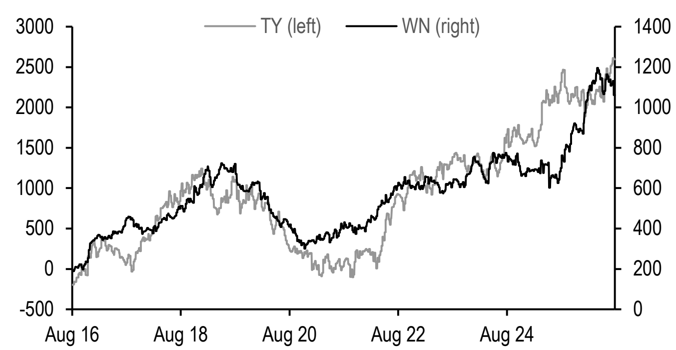
3-year statistics* on net asset manager longs for the various Treasury futures contracts; '000s of contracts, past 3Y history

Contract	Current	Min	Max	Mean	Std. Dev.	Percentile	Z-score
WN	1061	603	1196	794	169	83%	1.6
US	552	371	656	500	58	79%	0.9
UXY	713	184	747	473	161	96%	1.5
TY	2595	1117	2612	1782	412	99%	2.0
FV	2963	1598	3836	2831	661	58%	0.2
TU	1693	1128	2603	1875	376	28%	-0.5

Current as of 8/11/2026
Source: J.P. Morgan, CFTC

Figure 8: Positioning in TY is close to all-time highs, while positioning in WN has declined modestly in the last two weeks

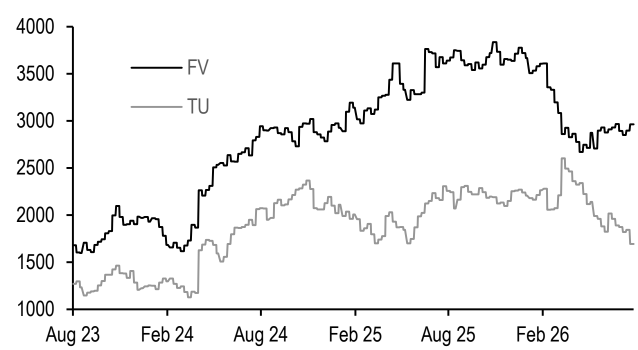
Net asset manager longs in the WN (right) and TY (left) sectors, past 10Y; 000s of contracts



Source: J.P. Morgan, CFTC

Figure 9: Asset manager longs in the shorter-dated contracts have declined this year

Net asset manager longs in the FV and TU sectors, past 3Y; 000s of contracts



Source: J.P. Morgan, CFTC

CTD switch optionality, basis convergence, and the wildcard option

In the **TU sector**, the CTD into the front contract is the 4.125% Jun '28; the 3.875% Jun '28 and 4% Jun '28 are near-equicheap. The CTD into the back contract is the 4.625% Sep '28, with the 3.375% Sep '28 equicheap. All of these bonds have a net basis of zero, so a CTD switch is unlikely to affect the futures price. Both contracts are fairly priced, so basis convergence should be a non-issue for TU calendar spreads this cycle.

In the **FV sector**, the CTD into the front contract is the 3.5% Nov '30, and the CTD into the back contract is the 3.5% Feb '31. A CTD switch is unlikely for either contract. Both contracts are fairly priced, and thus basis convergence should be a non-issue for FV calendar spreads this cycle.

In the **TY sector**, the CTD into the front contract is the 4.25% May '33, and a CTD switch is unlikely. For the back contract, the WI 4.5% Aug '33 is currently the CTD, with the 4.375% Jul '33 and 3.875% Aug '33 equicheap; the three have net bases within ~0.5/32nds, so a CTD switch is unlikely to affect the futures price. Currently, we value the delivery option value for the back contract at ~4+/32nds, which indicates that the back basis is underpricing the optionality, trading near flat. However, some of this optionality is due to the WI being CTD, and the TY back basis tends to remain stable through the roll period. Consequently, we don't expect basis convergence to have an

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impact on TY calendar spreads.

In the **UXY sector**, the CTD into the front contract is the 4.125% Feb '36, and the CTD into the back contract is the 4.375% May '36. CTD switches are unlikely, and with both bases trading near zero, basis convergence is a non-issue for UXY calendar spreads this cycle.

In the **classic bond contract (US)**, the CTDs for the front and back contracts are both the 5% May '45, and a number of bonds are nearly equicheap for each. The front contract is currently trading at a net basis of ~2+/32nds, and the back contract is trading at a net basis of ~4+/32nds. Both bases are not currently fully pricing in the delivery optionality, as we see the front contract delivery option value at ~4+/32nds and that for the back contract at ~15/32nds.

We wrote in detail about the switch option value in the US (and also WN) contract in *Born(e) in the US(A)*,7/31/2026. In summary, the recent selloff and steepening of the curve has resulted in increased optionality in the contract; if yields continue to sell off or if the curve continues to steepen, a CTD switch to a higher-duration bond could occur. Therefore, we view long basis positions in the back contract as an attractive way to position for higher yields. Additionally, some of the potential switches could be to lower-coupon bonds, which could make the wildcard option more material than it currently is, resulting in the basis widening.

Therefore, **with the back basis not fully pricing in the optionality and the attractiveness of long basis positions in the back contract, the back basis could widen, which could pressure calendar spreads wider**.

In the **WN contract**, the CTDs for both the front and back contracts are the 4% Nov '52. CTD switches are unlikely absent a significant steepening of the curve. The front contract is currently trading at a net basis of 2/32nds, and the back contract is currently trading at a net basis of 1+/32nds. The front basis is slightly overpricing the optionality, as we see the delivery option value at 1/32nds. In the back contract, the optionality seems to be underpriced, as we see a delivery option value of 3+/32nds for the back contract. Taken together, **basis convergence could exert a mildly bullish widening pressure on the calendar spreads, especially given the potential attractiveness of the back basis**.

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Contract tables

Table 1: Key attributes of the front and back Ultra-long bond futures contracts

Projected key attributes of the front and back WN contracts*

	Front	Back		Front	Back
Price	110-15	110-03	CTD carry (32nds)	3.8	9.5
DV01 ($/contract)	173.53	175.28	Repo sensitivity (32nds/10bp)	0.36	1.04
Hedge Ratio	1000	-990	Gross Basis (32nds)	5.8	11.1
CTD	4 Nov 52	4 Nov 52	BNOC (32nds)	2.0	1.6
CTD repo	3.69	3.86	First Trade Date	Dec 22 2025	Mar 23 2026
Switch option value (32nds)	0.8	3.8	Last Trade Date	Sep 21 2026	Dec 21 2026
Implied Repo	3.14	3.70	First Notice Date	Aug 31 2026	Nov 30 2026
			First Delivery Date	Sep 01 2026	Dec 01 2026
			Last Delivery Date	Sep 30 2026	Dec 31 2026
Calendar Spread	12/32nds				

Source: J.P. Morgan, Bloomberg Finance L.P., CME
* Market data as of COB 8/11/2026

Table 2: Key attributes of the front and back "Classic" bond futures contracts

Projected key attributes of the front and back US contracts*

	Front	Back		Front	Back
Price	109-03	108-20	CTD carry (32nds)	5.5	14.3
DV01 ($/contract)	146.90	149.90	Repo sensitivity (32nds/10bp)	0.43	1.23
Hedge Ratio	1000	-980	Gross Basis (32nds)	7.8	18.7
CTD	5 May 45	5 May 45	BNOC (32nds)	2.3	4.4
CTD repo	3.69	3.86	First Trade Date	Dec 22 2025	Mar 23 2026
Switch option value (32nds)	4.6	14.8	Last Trade Date	Sep 21 2026	Dec 21 2026
Implied Repo	3.16	3.50	First Notice Date	Aug 31 2026	Nov 30 2026
			First Delivery Date	Sep 01 2026	Dec 01 2026
			Last Delivery Date	Sep 30 2026	Dec 31 2026
Calendar Spread	15/32nds				

Source: J.P. Morgan, Bloomberg Finance L.P., CME
* Market data as of COB 8/11/2026

Table 3: Key attributes of the front and back Ultra 10-year note contracts

Projected key attributes of the front and back UXY contracts*

	Front	Back		Front	Back
Price	110-08	109-30+	CTD carry (32nds)	2.2	6.4
DV01 ($/contract)	85.83	87.80	Repo sensitivity (32nds/10bp)	0.42	1.24
Hedge Ratio	1000	-977	Gross Basis (32nds)	2.5	6.6
CTD	4-1/8 Feb 36	4-3/8 May 36	BNOC (32nds)	0.3	0.2
CTD repo	3.69	3.86	First Trade Date	Dec 22 2025	Mar 23 2026
Switch option value (32nds)	0.0	0.1	Last Trade Date	Sep 21 2026	Dec 21 2026
Implied Repo	3.61	3.85	First Notice Date	Aug 31 2026	Nov 30 2026
			First Delivery Date	Sep 01 2026	Dec 01 2026
			Last Delivery Date	Sep 30 2026	Dec 31 2026
Calendar Spread	9+/32nds				

Source: J.P. Morgan, Bloomberg Finance L.P, CME
* Market data as of COB 8/11/2026

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Table 4: Key attributes of the front and back 10-year T-note futures contracts

Projected key attributes of the front and back TY contracts*

	Front	Back		Front	Back
Price	108-15+	108-07+	CTD carry (32nds)	2.3	6.5
DV01 ($/contract)	64.43	67.11	Repo sensitivity (32nds/10bp)	0.43	1.08
Hedge Ratio	1000	-960	Gross Basis (32nds)	2.3	6.9
CTD	4-1/4 May 33	WI Aug 33	BNOC (32nds)	0.0	0.4
CTD repo	3.69	3.88	First Trade Date	Dec 22 2025	Mar 23 2026
Switch option value (32nds)	0.4	4.7	Last Trade Date	Sep 21 2026	Dec 21 2026
Implied Repo	3.69	3.84	First Notice Date	Aug 31 2026	Nov 30 2026
			First Delivery Date	Sep 01 2026	Dec 01 2026
			Last Delivery Date	Sep 30 2026	Dec 31 2026
Calendar Spread	8/32nds				

Source: J.P. Morgan, Bloomberg Finance L.P., CME
* Market data as of COB 8/11/2026

Table 5: Key attributes of the front and back 5-year T-note futures contracts

Projected key attributes of the front and back FV contracts*

	Front	Back		Front	Back
Price	106-10+	106-06+	CTD carry (32nds)	-0.3	-2.6
DV01 ($/contract)	42.03	45.33	Repo sensitivity (32nds/10bp)	0.17	0.96
Hedge Ratio	1000	-927	Gross Basis (32nds)	-0.3	-2.5
CTD	3-1/2 Nov 30	3-1/2 Feb 31	BNOC (32nds)	0.0	0.1
CTD repo	3.69	3.86	First Trade Date	Jan 02 2026	Apr 01 2026
Switch option value (32nds)	0.0	0.2	Last Trade Date	Sep 30 2026	Dec 31 2026
Implied Repo	3.69	3.85	First Notice Date	Aug 31 2026	Nov 30 2026
			First Delivery Date	Sep 01 2026	Dec 01 2026
			Last Delivery Date	Oct 05 2026	Jan 06 2027
Calendar Spread	4/32nds				

Source: J.P. Morgan, Bloomberg Finance L.P., CME
* Market data as of COB 8/11/2026

Table 6: Key attributes of the front and back 2-year T-note futures contracts

Projected key attributes of the front and back TU contracts*

	Front	Back		Front	Back
Price	102-30+	102-27	CTD carry (32nds)	1.6	8.6
DV01 ($/contract)	36.88	42.25	Repo sensitivity (32nds/10bp)	0.48	1.33
Hedge Ratio	1000	-872	Gross Basis (32nds)	1.5	8.5
CTD	4-1/8 Jun 28	4-5/8 Sep 28	BNOC (32nds)	-0.1	-0.1
CTD repo	3.69	3.86	First Trade Date	Jan 02 2026	Apr 01 2026
Switch option value (32nds)	0.0	0.0	Last Trade Date	Sep 30 2026	Dec 31 2026
Implied Repo	3.72	3.87	First Notice Date	Aug 31 2026	Nov 30 2026
			First Delivery Date	Sep 01 2026	Dec 01 2026
			Last Delivery Date	Oct 05 2026	Jan 06 2027
Calendar Spread	4/32nds				

Source: J.P. Morgan, Bloomberg Finance L.P., CME
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Appendix

1. *Calendar spread relative value.* For contracts with negligible embedded option value, the relative value of the calendar spread can be determined by comparing the implied repo rate (IRP) of the calendar spread to the forward three-month funding rate. The calendar spread IRP is the breakeven funding rate for going long the calendar spread, taking delivery in the front-month contract, and making delivery in the back-month contract. If the calendar IRP is historically low compared with the forward financing rate for a particular market, the calendar spread may be considered rich.

   For contracts for which delivery options in the futures are valuable, such as U.S. Treasury bond futures, we think a simple IRP analysis is inadequate because it overlooks the potential impact of switches in the cheapest-to-deliver bond. In these cases, the option-adjusted relative values of the front and back contracts should be compared to determine the relative value of the calendar spread. If the Option-Adjusted Basis Net of Carry (OABNOC) or Risk-Adjusted Yield Spread (RAYS) is lower for the front-month contract than for the back-month contract, the calendar spread is considered rich.
2. *Changes in short-term funding rates.* An increase in funding rates generally causes futures prices to rise relative to cash prices, since it decreases the carry basis. Thus, the calendar spread reflects changes in the shape of the short end of the yield curve. For example, if four-month funding rates rise while one-month rates remain the same, the calendar spread is expected to fall. Changes in funding rates affect back-month futures more than front-month futures, since the back-month contract has three months more carry in its basis. As a rule of thumb, for a parallel change in funding rates:

   -1 x (change in funding rate) / 4 = change in calendar spread.

   For example, if short-term rates increase by 25 basis points, the calendar spread is expected to decrease by approximately 0.0625 or 2/32nds:

   -1 x 0.25 / 4 = -0.0625 (in decimal form, or -2/32nds).
3. *Supply and demand pressures.* The calendar spread may become richer or cheaper depending on the balance of long and short hedgers and investors who are rolling positions.

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